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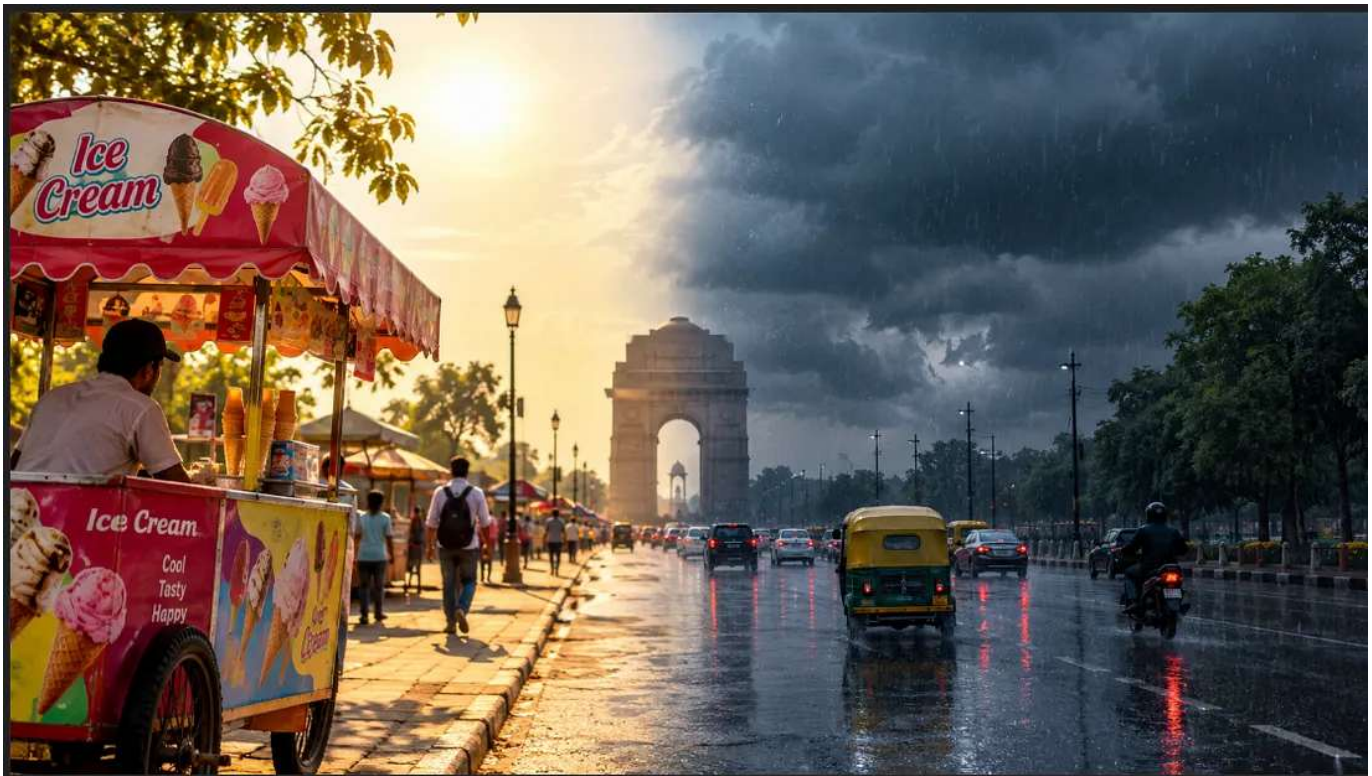
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Why Your Brain Is Wrong About "Cause and Effect" (An Indian Summer Story)



Tarun Chaudhary 5 min read · Just now



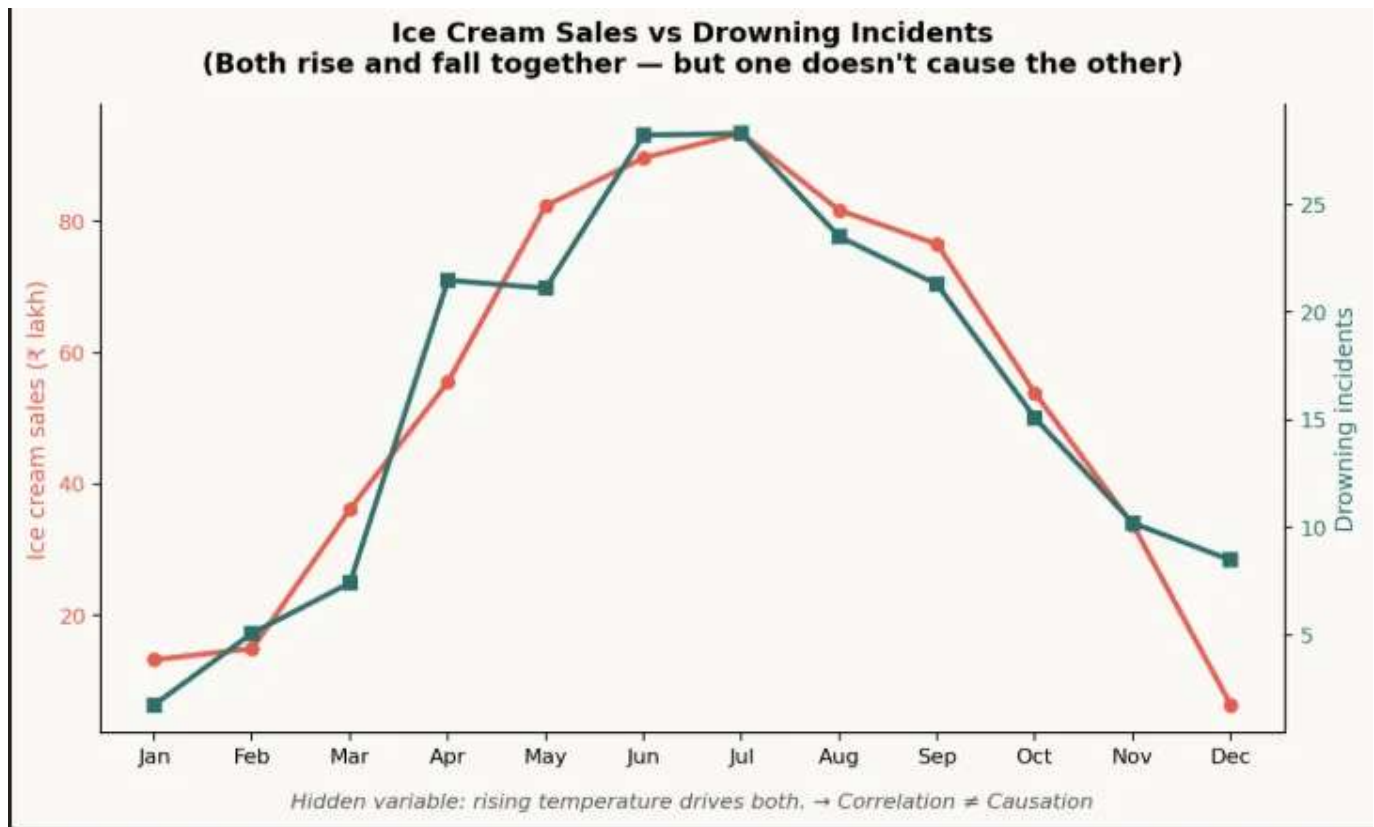
Every May, as Delhi's mercury climbs past 40°C, two things happen with clockwork regularity:

1. Air conditioner sales spike.
2. A few months later, dengue cases spike too.

Plot both on a graph and you'll get a relationship so clean it looks scripted — both curves rising together through summer, both falling together by November.

So does buying an AC give you dengue?

Obviously not. But if you only had the numbers and no context, you might be tempted to write that headline. This is the single most common mistake in data journalism, business dashboards, and let's be honest WhatsApp University forwards: **mistaking correlation for causation.**



The textbook example: ice cream sales and drowning deaths rise and fall together every year. Nobody actually believes ice cream causes drowning — because we all know the real driver: summer heat.

What correlation actually means

Correlation just means two things move together — when one goes up, the other tends to go up (or down) too. Statisticians measure this with a number between -1 and +1 called the correlation coefficient. Close to +1: they move together. Close to -1: they move in opposite directions. Close to 0: no relationship.

That's it. That's *all* correlation tells you. It says nothing about *why* they move together. It doesn't tell you which one is the cause, whether either one is the cause, or whether some third factor is quietly pulling both strings from behind the curtain.

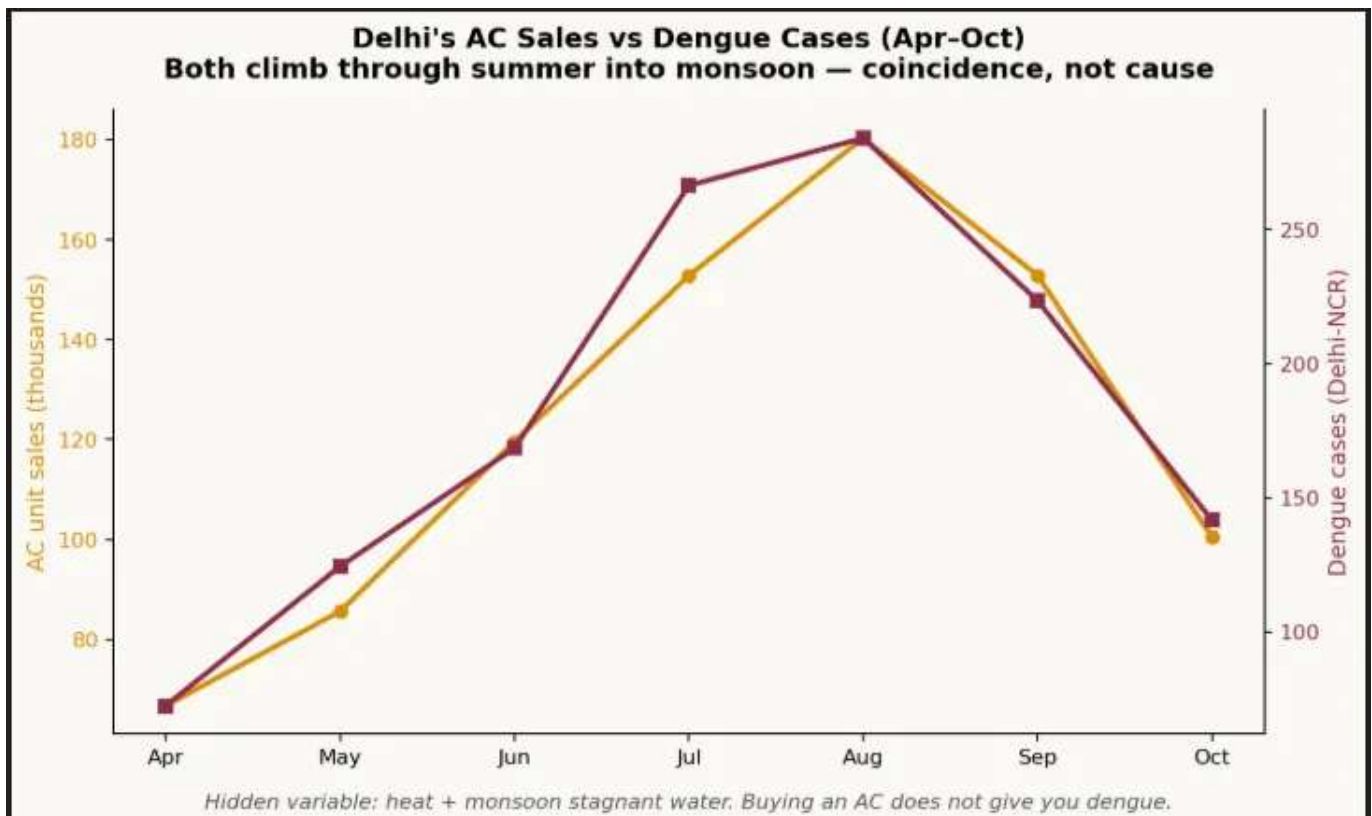
Causation is a much stronger claim: it says A *produces* B. Change A, and B changes as a direct result. Proving that requires far more than a matching

graph — it usually needs a controlled experiment, a natural experiment, or a mountain of ruled-out alternatives.

The Delhi AC-Dengue story, properly explained

Here's the real chain of events:

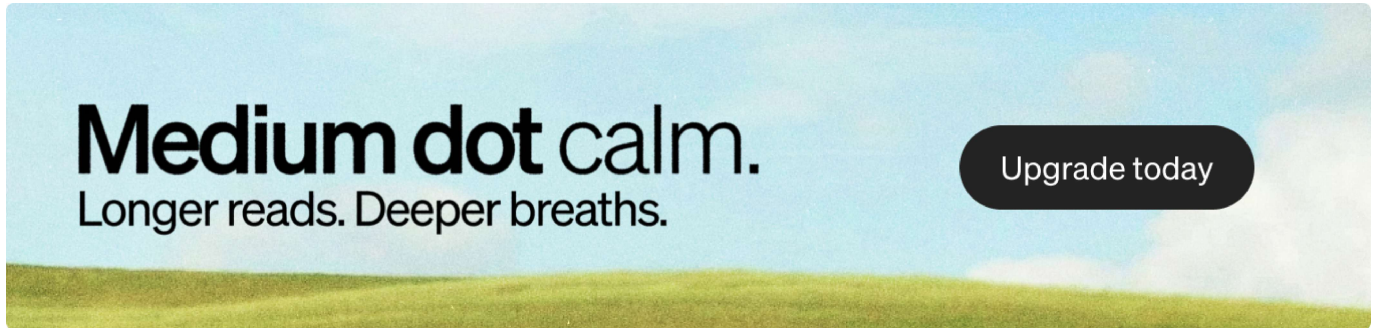
- Rising heat (April–June) → people buy ACs to survive.
- Rising heat + monsoon rains (July–September) → stagnant water collects in coolers, tyres, terraces, construction sites → *Aedes aegypti* mosquitoes breed → dengue cases climb.



AC sales and dengue cases both track the same underlying driver → heat followed by monsoon stagnant water not each other.

The AC and the dengue mosquito never met. They're both just downstream of the same weather pattern. In statistics, this shared, unaccounted-for

driver is called a **confounding variable** (or a **lurking variable**). Once you spot the confounder, the “spooky” correlation stops being spooky and becomes obvious.



This is the pattern behind almost every “shocking correlation” you’ll see shared online:

- States with more temples also report higher crime numbers → confounder: population. More people, more of everything — temples *and* crime.
- Areas near railway stations sell more paan and also have higher pickpocketing rates → confounder: footfall.
- IPL viewership and electricity consumption both spike on match nights → confounder: hundreds of millions of TVs and ACs switched on at the same hour — the match isn’t “causing” the grid to strain on its own; the shared trigger is simultaneous behaviour, not one causing the other in a production sense.

Why our brains fall for this trap so easily

Humans are pattern-seeking by default — it’s an evolutionary shortcut that once helped us survive (“rustle in the grass + tiger last time = rustle means

tiger"). But that same shortcut misfires constantly with data, because:

- **We crave simple stories.** "AC sales cause dengue" is a punchier headline than "a confluence of temperature and monsoon hydrology increases vector-borne disease incidence."
- **We rarely check for a third variable.** Two-variable charts feel complete. They're not.
- **Timing fools us.** If A happens before B, we assume A caused B. This is a specific, well-documented fallacy called *post hoc ergo propter hoc* — "after this, therefore because of this." A rooster crows before sunrise every single day. The rooster does not cause the sun to rise.
- **Reverse causation gets ignored.** Sometimes it's not "A causes B" but "B causes A." Cities with more police stations tend to report more crime — not because police create crime, but because more reporting infrastructure captures more of the crime that was already happening.

Where this actually costs money and lives

This isn't just a fun statistics party trick getting it wrong has real consequences:

- **Business:** A D2C startup notices that customers who open push notifications 5+ times a week have 3x higher retention, so they start spamming everyone with notifications. Retention doesn't budge. Why? Engaged customers who already love the product were opening notifications *because* they were engaged — the notifications weren't creating the engagement.
- **Public health:** Early observational studies linked moderate wine drinking to a lower heart-disease risk. Turned out people who drink moderately

also tend to be wealthier, more health-conscious, and have better healthcare access. The wine wasn't the hero — the confounders were.

- **Policy:** A state sees literacy rates and per-capita income rise together and assumes literacy programs alone drove economic growth, when in reality both were driven by decades of infrastructure investment, urbanization, and industrial policy acting together.

Treating a correlation as a lever you can pull is how companies waste marketing budgets and how governments waste public money on interventions that quietly do nothing.

So how do you actually prove causation?

You need one of these, roughly in order of rigor:

1. **Randomized Controlled Trials (RCTs):** Randomly assign subjects to a treatment group and a control group. Randomization is what neutralizes confounders — since assignment is random, both groups should be statistically identical on every other factor. This is the gold standard, used in drug trials and (increasingly) in product A/B tests.
2. **Natural experiments:** When you can't randomize, look for situations where something external did the randomizing for you — like a policy change that applied to one state but not a neighboring one with similar demographics.
3. **Instrumental variables & regression discontinuity:** Statistical techniques economists use to isolate causal effects from observational data when RCTs aren't feasible.
4. **Granger causality tests:** Useful for time-series data — checking whether past values of A actually improve predictions of future B, beyond what B's

own past already tells you. (Necessary but not sufficient for true causation.)

5. Mechanism, replication, and ruling out alternatives: Even without a formal experiment, a causal claim gets stronger when you can explain a plausible mechanism, replicate the finding across different populations, and systematically rule out known confounders.

None of these are as easy as "look, the lines match." That's exactly the point real causal claims should be harder to make than that.

The one-line filter to apply before you believe any "X causes Y" claim

Ask: what third factor could be driving both of these at once — and has anyone actually tried to remove its influence?

If the answer is "nobody checked," you're not looking at causation. You're looking at two lines on a chart that happened to be in the same room together.

Next monsoon, when your cousin forwards a message claiming "states with more mobile towers have higher crime rates, proof that 5G causes crime" — you'll know exactly what to say back.

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